Definition 2.2.1. An **associative algebra** over k is a vector space A over k together with a bilinear map $A \times A \to A$, $(a, b) \mapsto ab$, such that $(ab)c = a(bc)$.

Definition 2.2.2. A **unit** in an associative algebra A is an element $1 \in A$ such that $1a = a1 = a$.

Proposition 2.2.3. *If a unit exists, it is unique.*

Proof. Let $1, 1'$ be two units. Then $1 = 11' = 1'$. □

From now on, by an algebra A we will mean an associative algebra with a unit.

Example 2.2.4. Here are some examples of algebras over k:

1. $A = k$.

2. $A = k[x_1, \ldots, x_n]$ — the algebra of polynomials in variables $x_1, \ldots, x_n$.

3. $A = \mathrm{End}V$ — the algebra of endomorphisms of a vector space V over k (i.e., linear maps, or operators, from V to itself). The multiplication is given by composition of operators.

4. The **free algebra** $A = k\langle x_1, \ldots, x_n\rangle$. A basis of this algebra consists of words in letters $x_1, \ldots, x_n$, and multiplication in this basis is simply the concatenation of words.

5. The **group algebra** $A = k[G]$ of a group G. Its basis is $\{a_g, g \in G\}$, with multiplication law $a_g a_h = a_{gh}$.

Definition 2.2.5. An algebra A is **commutative** if $ab = ba$ for all $a, b \in A$.

For instance, in the above examples, A is commutative in cases 1 and 2 but not commutative in cases 3 (if $\dim V > 1$) and 4 (if $n > 1$). In case 5, A is commutative if and only if G is commutative.

Definition 2.2.6. A **homomorphism of algebras** $f : A \to B$ is a linear map such that $f(xy) = f(x)f(y)$ for all $x, y \in A$ and $f(1) = 1$.